



JURONG JUNIOR COLLEGE
2017 JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

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CLASS

17S

EXAM INDEX

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CHEMISTRY

9729/04

Paper 4 Practical

17 August 2017
2 hours 30 minutes

Candidates answer on the Question paper.

Additional Materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 19 and 20.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	13
2	16
3	20
4	6
Total	55

This document consists of **20** printed pages and **2** blank pages.

Answer **all** the questions in the spaces provided.

For
Examiner's
Use

1 Determination of the percentage by mass of iron(II) sulfate in mixture

FA 1 is an aqueous solution containing a mixture of iron(II) sulfate, FeSO_4 and iron(III) sulfate, $\text{Fe}_2(\text{SO}_4)_3$.

In the presence of acid, the iron(II) sulfate is oxidised by potassium manganate(VII).



FA 1 is an aqueous solution containing 215.0 g dm^{-3} of a mixture of iron(II) sulfate, FeSO_4 and iron(III) sulfate, $\text{Fe}_2(\text{SO}_4)_3$.

FA 2 is 2.00 g dm^{-3} potassium manganate(VII), KMnO_4 .

FA 3 is 1.00 mol dm^{-3} sulfuric acid, H_2SO_4 .

In this question, you will perform a dilution followed by a titration. The data from this titration will be used to determine:

- the concentration of FeSO_4 in **FA 1**,
- the percentage by mass of FeSO_4 in the mixture used to prepare **FA 1**.

(a) (i) Dilution of FA 1

- Pipette 25.0 cm^3 of **FA 1** into the 250 cm^3 volumetric flask.
- Make up the contents of the flask to the 250 cm^3 mark with deionised water.
- Place the stopper in the flask and mix the contents thoroughly by slowly inverting the flask a number of times.
- Label this diluted solution of iron(II) sulfate and iron(III) sulfate **FA 4**.

Titration of FA 4 against FA 2

- Fill a burette with **FA 2**.
- Rinse the pipette thoroughly with deionised water and then with a little **FA 4**.
- Pipette 25.0 cm^3 of **FA 4** into the conical flask.
- Using an appropriate measuring cylinder, transfer 10.0 cm^3 of **FA 3** to the same conical flask.
- Run **FA 2** from the burette into this flask until the first **permanent** pale pink colour is obtained.
- Record your titration results in the space provided on page 3. Make certain that your recorded results show the precision of your working.
- Repeat points 1 to 6 as necessary until consistent results are obtained.

Results

For
Examiner's
Use

[6]

- (ii) From your titrations, obtain a suitable volume of **FA 2** to be used in your calculations. Show clearly how you obtained this volume.

Volume of **FA 2** = [1]

- (b) (i) Calculate the amount of KMnO_4 present in the volume calculated in (a)(ii).

[A_r : O, 16.0; K, 39.1; Mn, 54.9]

Amount of KMnO_4 = [1]

- (ii) Calculate the concentration, in mol dm^{-3} , of FeSO_4 in **FA 4**.

Concentration of FeSO_4 in **FA 4** = [1]

- (iii) Calculate the concentration, in mol dm^{-3} , of FeSO_4 in **FA 1**.

Concentration of FeSO_4 in **FA 1** = [1]

- (iv) **FA 1** is an aqueous solution containing 215.0 g dm^{-3} of a mixture of FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$.

Hence, calculate the percentage, by mass, of FeSO_4 in this mixture.

[Ar: O, 16.0; S, 32.1; Fe, 55.8]

Percentage by mass of FeSO_4 in mixture = [1]

- (c) The maximum error for a 25.0 cm^3 pipette commonly used in schools is $\pm 0.06 \text{ cm}^3$.

The maximum error in any single burette reading is $\pm 0.05 \text{ cm}^3$.

Calculate the maximum percentage error in each of the following.

- (i) The volume of **FA 4** pipetted into the conical flask.

Maximum % error =

- (ii) The volume of **FA 2** used in (a)(ii).

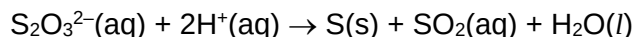
Maximum % error = [2]

[Total: 13]

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- 2 In this experiment you will investigate how the rate of reaction between sodium thiosulfate and sulfuric acid is affected by the concentration of sodium thiosulfate.

When aqueous thiosulfate ions react with hydrogen ions, H^+ , in any acid, a pale yellow precipitate of sulfur is formed. The ionic equation for this reaction is given below.

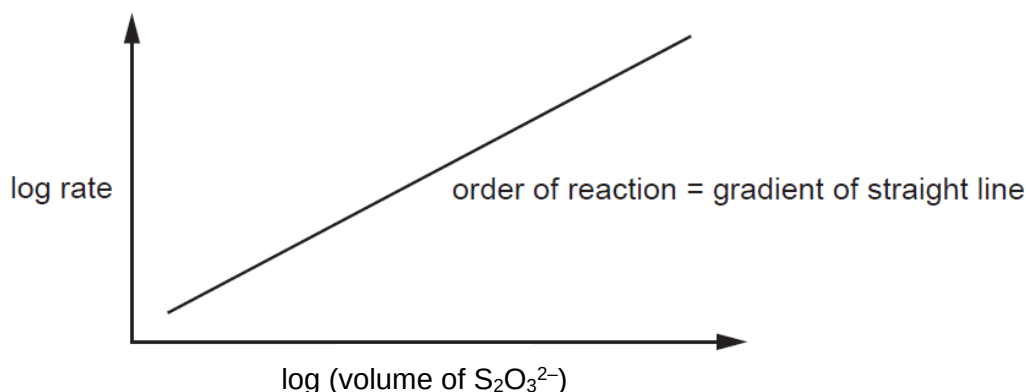


The rate of the reaction can be determined by measuring the time taken to produce a fixed quantity of sulfur.

FA 5 is 37.0 g dm^{-3} hydrated sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$

You will need access to the **FA 3** solution you used earlier.

The order of reaction with respect to thiosulfate ions can be obtained by plotting a graph of log rate against log (volume of $\text{S}_2\text{O}_3^{2-}$).



You are advised to read the instructions before starting any practical work.

(a) Method

Experiment 1

- Use an appropriate measuring cylinder, transfer 50.0 cm^3 of **FA 5** into a 250 cm^3 beaker.
- Measure 5.0 cm^3 of **FA 3** in an appropriate measuring cylinder.
- Tip the acid **FA 3** from the measuring cylinder into the beaker and start timing **immediately**.
- Swirl the beaker to mix the reagents and place the beaker on top of the printed insert page provided. (Detach the last page)
- Look down through the solution in the beaker at the print on the insert
- Stop timing as soon as the precipitate of sulfur makes the print on the insert invisible.
- Record the time taken, t , to the **nearest second**.
- Empty and rinse the 250 cm^3 beaker.
- Dry the outside of the beaker ready for Experiment 2.

Experiment 2

- Use the measuring cylinder for **FA 5** in Experiment 1 to transfer 10.0 cm^3 of **FA 5** into the rinsed and dried beaker.
- Use the same measuring cylinder for **FA 5** to transfer 40.0 cm^3 of deionised water into the same beaker.
- Measure 5.0 cm^3 of **FA 3** in an appropriate measuring cylinder.
- Tip the acid **FA 3** from the measuring cylinder into the beaker and start timing **immediately**.
- Swirl the beaker to mix the reagents and place the beaker on top of the printed insert page provided.
- Look down through the solution in the beaker at the print on the insert.
- Stop timing as soon as the precipitate of sulfur makes the print on the insert invisible.
- Record the time taken, t , to the **nearest second**.
- Empty and rinse the 250 cm^3 beaker.
- Dry the outside of the beaker ready for Experiment 3.

Experiments 3 – 5

Carry out three more experiments to investigate the order of reaction with respect to thiosulfate ions. The volume of **FA 5** should not be less than those in Experiment 2. Deionised water is added to keep the total volume of the final reaction mixture constant.

When you have completed all of your experiments, empty and rinse the beaker.

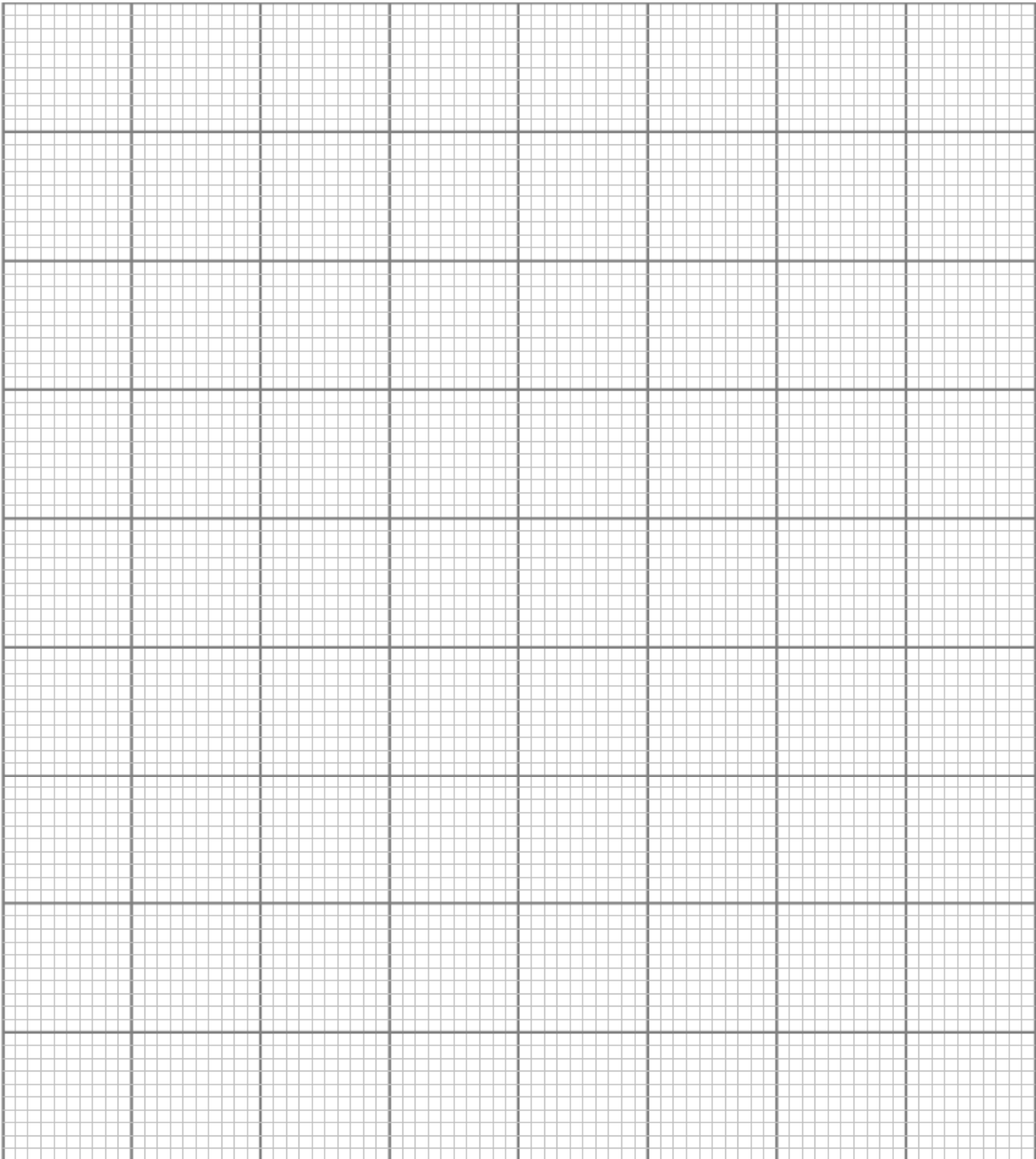
Prepare a table in the space provided to record, for each experiment

- all volumes
- the value of t , to the nearest second
- calculated value of $\log (1/t)$ and $\log (\text{volume of FA 5})$ respectively, to 3 significant figures

Results:

- (b) Use the grid below to plot a graph of **log (1/time)** against **log (volume of FA 5)**.
Draw an appropriate straight line through the points plotted.

For
Examiner's
Use



[3]

- (c) Draw construction lines on your graph and obtain data from them to enable you to calculate the gradient of the graph.

Hence, determine the order of reaction with respect to thiosulfate ions.

The reaction is order with respect to $\text{S}_2\text{O}_3^{2-}$. [5]

- (d) The experimental method can be modified to enable the rate of reaction with respect to hydrogen ions, H^+ , to be investigated.

In the first line of the table below, the volumes of **FA 5**, deionised water and **FA 3** used in **Experiment 2** are recorded.

Complete the following table, suggesting volumes for each of the reagents that could be used in two further experiments to investigate how the rate of reaction varies with a change in volume of sulfuric acid, **FA 3**.

Do not carry out this experiment.

Experiment	Volume of FA 5 / cm^3	Volume of deionised water/ cm^3	Volume of FA 3 / cm^3
2	10.0	40.0	5.0
(i)			
(ii)			

[1]

- (e) A student concluded that the reaction is first order with respect to hydrogen ions, H^+ .

Predict how the reaction time measured in **Experiment 2** would have been affected if the following modifications are made to the experiment separately.

For each case, explain your answer.

- (i) The experiment had been carried out using 1.0 mol dm^{-3} hydrochloric acid instead of 1.0 mol dm^{-3} sulfuric acid. [1]

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- (ii) The experiment had been carried out in a 100 cm^3 beaker instead of a 250 cm^3 beaker. [1]

.....

[Total: 16]

3 Qualitative Analysis

At each stage of any test, you are to record details of the following

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitate in an excess of the reagent added.

Where gases are released they should be identified by a test, **described in the appropriate place in your observations.**

You should indicate clearly at what stage in a test a change occurs.

No additional tests for ions present should be attempted.

If any solution is warmed, a boiling tube MUST be used.

Rinse and reuse test-tubes and boiling tubes where possible.

Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.

(a) **FA 6** is a solid and **FA 7** is a solution. **Some** of the ions in **FA 6** and **FA 7** are listed in the Qualitative Analysis notes on pages 19 and 20.

- (i) Place a **small** spatula measure of **FA 6** into a **hard-glass** test-tube.
Heat gently at first and then strongly, until no further change is observed.
Test with litmus papers while you are heating.
Record all your observations below.

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[2]

- (ii) Place two spatula measures of **FA 6** into a boiling-tube. Dissolve the solid in about 5 cm depth of distilled water. Use this **FA 6** solution together with the **FA 7** solution provided for the following tests.

test	observations	
	FA 6	FA 7
1. To a 0.5 cm depth of solution in a boiling tube add aqueous sodium hydroxide, then		
Warm gently.		
Allow to cool, add a piece of aluminium foil and warm again.	X	
2. To a 1 cm depth of solution in at test-tube add a 1 cm depth of dilute sulfuric acid followed by a few drops of aqueous potassium manganate(VII).		
3. To a 1 cm depth of solution in at test-tube add a 2 cm depth of hydrogen peroxide and leave to stand.		
4. To a 1 cm depth of solution in at test-tube add a 1 cm depth of dilute hydrochloric acid, then		
Add a 1 cm depth of aqueous barium nitrate.		

- (b) (i) Identify as many ions present in **FA 6** and **FA 7** as possible from your observations.

If an ion cannot be identified from the tests, write 'unknown' in the space.

	cation(s)	anion(s)
FA 6		
FA 7		

[2]

- (ii) Describe another test you could carry out to confirm the identity of a cation you have identified in (i). Record the reagent(s) and expected observation(s) in the space below.

Do not carry out this test.

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[2]

- (iii) Write an ionic equation for the reaction that would occur in (ii). Include state symbols.

.....

[1]

(c) Planning

Consider the following organic compounds.

benzaldehyde 2-methylpropan-2-ol propan-2-ol propanone

Plan an investigation, using test-tube reactions, **other** than using 2,4-dinitrophenylhydrazine reagent which would allow you to identify each of these four organic compounds.

Each compound should be identified by at least one positive test result. It is **not** sufficient to identify a compound simply by eliminating all the others.

Your plan should include:

- details of the reagents and conditions to be used,
- an outline of the sequence of steps you would follow,
- an explanation of how you would analyse your results in order to identify each compound.

Once a compound has been clearly identified, your plan should concentrate on distinguishing the remaining compounds.

[illegible]

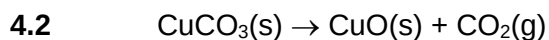
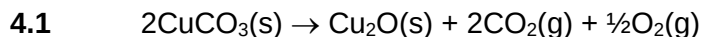
This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[6]

[Total: 20]

4. Planning

Copper has two oxides, Cu_2O and CuO . Copper(II) carbonate, CuCO_3 , decomposes on heating to form one of these oxides. Separate equations can be written showing the two possible decompositions.



You are to plan an experiment to investigate the decomposition of copper(II) carbonate on heating and hence decide which of the two equations represents the actual decomposition by collection of gas in the experiment.

- (a)** Draw a diagram of the apparatus and experimental set up you would use in the experiment.

Your apparatus should use only standard items found in a school or college laboratory and should show clearly

- how the copper(II) carbonate will be heated,
- how the volume of the gas evolved will be collected and measured.

Label each piece of apparatus used, indicating its size or capacity.

- (b) You are required to write a plan to determine which equation represents the actual decomposition of copper(II) carbonate

You may assume that you are provided with a sample of copper(II) carbonate

Your plan should include:

- calculation of the mass of copper(II) carbonate that will produce the maximum volume of gas that can be collected based on the capacity of the apparatus you have indicated in (a) for each equation when you are heating the solid.

(You may assume that the molar gas volume at the temperature in the apparatus used for gas collection stays constant at 25.3 dm^3 throughout the heating process.)

- practical details of how you would
 - determine the mass of copper(II) carbonate used in the experiment;
 - ensure the completion of the decomposition;
 - determine the volume of gas that has been collected.
- brief, but specific, details of how the results would then be used to reach a conclusion.

[Ar : C, 12.0; O, 16.0; Cu, 63.5]

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[illegible]

[Total: 6]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of Aqueous Cations

cation	Reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no. ppt.
chromium(III), Cr ³⁺ (aq)	grey–green ppt. soluble in excess giving dark green solution	grey–green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt. turning brown on contact with air insoluble in excess	green ppt. turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red–brown ppt. insoluble in excess	red–brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off–white ppt. rapidly turning brown on contact with air insoluble in excess	off–white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of Aqueous Anions

Anion	Reaction
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$);
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$);
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$);
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated on warming with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in excess dilute strong acids)

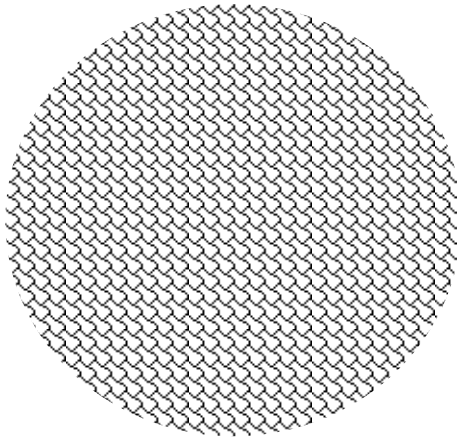
(c) Tests for Gases

gas	Test and test results
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns acidified aqueous potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	colour of element	colour in aqueous solution	colour in hexane
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas/liquid	orange	orange-red
iodine, I_2	black solid/purple gas	brown	purple

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